

Econometrics II (Panel Data) Problem Set

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Exercise 1: First Differences & Within Groups

(a) First Differences vs. Within Groups

Consider the static linear panel data model:

$$y_{it} = \beta x_{it} + \eta_i + v_{it} \quad (1)$$

where x_{it} is a scalar endogenous regressor and z_{it} is an $r \times 1$ instrument vector. Under the assumption of **strict exogeneity** [1]:

$$E[v_{it} | \eta_i, z_{i1}, \dots, z_{iT}] = 0 \quad (2)$$

We define the $T \times 1$ level error vector as $v_i = (v_{i1}, \dots, v_{iT})'$ and the $T \times r$ instrument matrix as $z_i = (z_{i1}, \dots, z_{iT})'$ [1].

2SLS Estimator Formulas

The 2SLS estimators for the transformed equations are defined as follows:

(i) First-Difference (FD) Estimator: Using Δz_{it} as the instrument for the equation $\Delta y_{it} = \beta \Delta x_{it} + \Delta v_{it}$:

$$\hat{\beta}_{FD} = \left(\left[\sum_{i=1}^N \Delta x_i' \Delta z_i \right] \left[\sum_{i=1}^N \Delta z_i' \Delta z_i \right]^{-1} \left[\sum_{i=1}^N \Delta z_i' \Delta x_i \right] \right)^{-1} \left[\sum_{i=1}^N \Delta x_i' \Delta z_i \right] \left[\sum_{i=1}^N \Delta z_i' \Delta z_i \right]^{-1} \left[\sum_{i=1}^N \Delta z_i' \Delta y_i \right] \quad (3)$$

where Δx_i and Δy_i are $(T-1) \times 1$ vectors, and Δz_i is a $(T-1) \times r$ matrix [2].

(ii) Within-Group (WG) Estimator: Using $\tilde{z}_{it} = z_{it} - \bar{z}_i$ as the instrument for the equation $\tilde{y}_{it} = \beta \tilde{x}_{it} + \tilde{v}_{it}$:

$$\hat{\beta}_{WG} = \left(\left[\sum_{i=1}^N \tilde{x}_i' \tilde{z}_i \right] \left[\sum_{i=1}^N \tilde{z}_i' \tilde{z}_i \right]^{-1} \left[\sum_{i=1}^N \tilde{z}_i' \tilde{x}_i \right] \right)^{-1} \left[\sum_{i=1}^N \tilde{x}_i' \tilde{z}_i \right] \left[\sum_{i=1}^N \tilde{z}_i' \tilde{z}_i \right]^{-1} \left[\sum_{i=1}^N \tilde{z}_i' \tilde{y}_i \right] \quad (4)$$

where $\tilde{x}_i$ and $\tilde{y}_i$ are $T \times 1$ vectors, and $\tilde{z}_i$ is a $T \times r$ matrix [2].

Asymptotic Distribution

To get the asymptotic distribution, first we can rewrite the estimators to depend on the level errors. Let D be the $(T-1) \times T$ first-difference operator and $Q = I_T - \frac{1}{T}\mathbf{1}\mathbf{1}'$ be the $T \times T$ within-group operator [3, 4]. Note that $\Delta v_i = Dv_i$ and $\tilde{v}_i = Qv_i$.

$$\hat{\beta}_{FD} = \beta + \left(\hat{S}_{xz}^{FD} (\hat{S}_{zz}^{FD})^{-1} \hat{S}_{zx}^{FD} \right)^{-1} \hat{S}_{xz}^{FD} (\hat{S}_{zz}^{FD})^{-1} \left[\frac{1}{N} \sum_{i=1}^N \Delta z_i' Dv_i \right] \quad (5)$$

$$\hat{\beta}_{WG} = \beta + \left(\hat{S}_{xz}^{WG} (\hat{S}_{zz}^{WG})^{-1} \hat{S}_{zx}^{WG} \right)^{-1} \hat{S}_{xz}^{WG} (\hat{S}_{zz}^{WG})^{-1} \left[\frac{1}{N} \sum_{i=1}^N \tilde{z}_i' Qv_i \right] \quad (6)$$

Both estimators explicitly depend on the level error vector v_i through the linear transformations D and Q .

As $N \rightarrow \infty$ with T fixed, the sample moments converge to their population counterparts (for example, $S_{xz}^{FD} = E[\hat{S}_{xz}^{FD}] = E[\Delta x_i' \Delta z_i]$ and $S_{zz}^{FD} = E[\hat{S}_{zz}^{FD}] = E[\Delta z_i' \Delta z_i]$ for the FD model). By the Law of Large Numbers (LLN) and the strict exogeneity assumption:

$$E[\Delta z_i' Dv_i] = E[E[\Delta z_i' Dv_i | z_i, \eta_i]] = E[\Delta z_i' D E[v_i | z_i, \eta_i]] = 0 \quad (7)$$

Similarly, $E[\tilde{z}_i' Qv_i] = 0$. Therefore, the expected limit values are:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{FD} = \beta \quad \text{and} \quad \text{plim}_{N \rightarrow \infty} \hat{\beta}_{WG} = \beta \quad (8)$$

This confirms that both estimators are consistent for β and gives us the expected value of the limiting distribution of $\sqrt{N}(\hat{\beta} - \beta)$ in both cases. Because the estimators are asymptotically normal, now we only need to derive the asymptotic variance V :

$$\begin{aligned} V &= \text{AVar} \left(\sqrt{N}(\hat{\beta} - \beta) \right) \\ &= (S_{xz} S_{zz}^{-1} S_{zx})^{-1} S_{xz} S_{zz}^{-1} \Omega S_{zz}^{-1} S_{zx} (S_{xz} S_{zz}^{-1} S_{zx})^{-1} \end{aligned}$$

Where Ω depends on the conditional covariance matrix $\Sigma(z_i) = \text{Var}(v_i | z_i)$. By the Law of Iterated Expectations (LIE), the unconditional variance of the scaled score is obtained by taking the expected value of the conditional variance. Specifically:

$$\Omega_{FD} = \text{Var}(\Delta z_i' \Delta v_i) = E[\Delta z_i' \Delta v_i \Delta v_i' \Delta z_i] = E[\Delta z_i' E[\Delta v_i \Delta v_i' | z_i] \Delta z_i] = E[\Delta z_i' D \Sigma(z_i) D' \Delta z_i]$$

$$\Omega_{WG} = \text{Var}(\tilde{z}_i' \tilde{v}_i) = E[\tilde{z}_i' \tilde{v}_i \tilde{v}_i' \tilde{z}_i] = E[\tilde{z}_i' E[\tilde{v}_i \tilde{v}_i' | z_i] \tilde{z}_i] = E[\tilde{z}_i' Q \Sigma(z_i) Q' \tilde{z}_i]$$

This gives us the formula for V that depends on Ω . Yet, we still need to compute $\Sigma(z_i)$, as in this particular case the persistency of the error term implies a specific structure for the covariance matrix.

The error process is given by $v_{i1} \sim (0, \sigma^2)$ and $v_{it} = \rho v_{i,t-1} + \epsilon_{it}$ with $\text{Var}(\epsilon_{it} | z_i) = \sigma^2$ [1]. To convincingly show how this process dictates the structure of $\Sigma(z_i)$, we can write a few steps of the iterative backward expansion of the AR(1) process:

$$\begin{aligned} v_{it} &= \rho v_{i,t-1} + \epsilon_{it} \\ &= \rho(\rho v_{i,t-2} + \epsilon_{i,t-1}) + \epsilon_{it} \\ &= \rho^2 v_{i,t-2} + \rho \epsilon_{i,t-1} + \epsilon_{it} \\ &\vdots \\ &= \rho^{t-1} v_{i1} + \sum_{j=0}^{t-2} \rho^j \epsilon_{i,t-j} \end{aligned}$$

Because the initial value v_{i1} and the innovations ϵ_{it} are mutually orthogonal and serially uncorrelated conditionally on z_i , the conditional variance corresponds to the sum of the variances. Therefore, the elements $\sigma_{ts} = \text{Cov}(v_{it}, v_{is}|z_i)$ of the $T \times T$ matrix $\Sigma(z_i)$ are derived as follows:

1. Diagonal Elements (Variances):

$$\text{Var}(v_{it}|z_i) = \rho^{2(t-1)}\text{Var}(v_{i1}|z_i) + \sum_{j=0}^{t-2} \rho^{2j}\text{Var}(\epsilon_{i,t-j}|z_i) = \sigma^2 \sum_{j=0}^{t-1} \rho^{2j} = \sigma^2 \frac{1 - \rho^{2t}}{1 - \rho^2} \quad (9)$$

2. Off-Diagonal Elements (Covariances): For $t > s$, we can express the error as $v_{it} = \rho^{t-s}v_{is} + \sum_{j=0}^{t-s-1} \rho^j \epsilon_{i,t-j}$. Taking the conditional covariance yields:

$$\text{Cov}(v_{it}, v_{is}|z_i) = E \left[\left(\rho^{t-s}v_{is} + \sum_{j=0}^{t-s-1} \rho^j \epsilon_{i,t-j} \right) v_{is} \middle| z_i \right] = \rho^{t-s}\text{Var}(v_{is}|z_i) = \rho^{t-s}\sigma^2 \frac{1 - \rho^{2s}}{1 - \rho^2} \quad (10)$$

This matrix $\Sigma(z_i)$ natively characterizes the serial correlation and heteroskedasticity that ultimately determine the relative efficiency of the FD and WG estimators.

(b) Unfeasible Optimal IV Estimator and Efficiency

To solve part (b), we derive the unfeasible optimal IV estimator and its asymptotic variance by treating the problem as Optimal GLS in the differenced system. We then show under which parameter values the standard First-Difference (FD) and Within-Group (WG) 2SLS estimators achieve this optimal efficiency.

Unfeasible Optimal IV Estimator

We eliminate the time-invariant unobserved heterogeneity η_i by applying a $(T-1) \times T$ full-row rank transformation matrix K (e.g., the first-difference matrix D) to the stacked level equations:

$$Ky_i = Kx_i\beta + Kv_i \quad (11)$$

By the strict exogeneity of the instruments, $E[Kv_i | z_i] = 0$. The conditional variance of the transformed errors is:

$$\Omega(z_i) \equiv \text{Var}(Kv_i | z_i) = K\Sigma(z_i)K' \quad (12)$$

where $\Sigma(z_i) = \text{Var}(v_i | z_i)$ is parametrized by ρ and σ^2 .

We are given the linear first-stage assumption in levels: $E[x_i | z_i] = z_i\Pi$. Consequently, the optimal instrument for the transformed endogenous regressor is its conditional expectation:

$$E[Kx_i | z_i] = Kz_i\Pi \quad (13)$$

As shown for the Optimal GLS estimator, the most efficient estimator weights the observations by the inverse of the conditional variance of the transformed errors, $\Omega(z_i)^{-1} = (K\Sigma(z_i)K')^{-1}$. Using the optimal instruments and this optimal weighting matrix, the unfeasible optimal IV estimator is:

$$\hat{\beta}_{IV} = \left(\sum_{i=1}^N (Kz_i\Pi)' (K\Sigma(z_i)K')^{-1} (Kx_i) \right)^{-1} \left(\sum_{i=1}^N (Kz_i\Pi)' (K\Sigma(z_i)K')^{-1} (Ky_i) \right) \quad (14)$$

Asymptotic Variance

To find the asymptotic distribution, we substitute $Ky_i = Kx_i\beta + Kv_i$ into the estimator:

$$\sqrt{N}(\hat{\beta}_{IV} - \beta) = \left(\frac{1}{N} \sum_{i=1}^N \Pi' z_i' K' (K\Sigma(z_i)K')^{-1} Kx_i \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \Pi' z_i' K' (K\Sigma(z_i)K')^{-1} Kv_i \right) \quad (15)$$

By the LLN and the LIE, the probability limit of the Hessian term is:

$$E [\Pi' z_i' K' (K\Sigma(z_i)K')^{-1} E[Kx_i | z_i]] = E [\Pi' z_i' K' (K\Sigma(z_i)K')^{-1} Kz_i \Pi] \quad (16)$$

By the Central Limit Theorem, the score vector converges to a normal distribution with variance:

$$\begin{aligned} \text{Var} (\Pi' z_i' K' (K\Sigma(z_i)K')^{-1} Kv_i) &= E [\Pi' z_i' K' (K\Sigma(z_i)K')^{-1} E[Kv_i v_i' K' | z_i] (K\Sigma(z_i)K')^{-1} Kz_i \Pi] \\ &= E [\Pi' z_i' K' (K\Sigma(z_i)K')^{-1} (K\Sigma(z_i)K') (K\Sigma(z_i)K')^{-1} Kz_i \Pi] \\ &= E [\Pi' z_i' K' (K\Sigma(z_i)K')^{-1} Kz_i \Pi] \end{aligned}$$

Because the expected Hessian and the variance of the score are identical (Information Matrix Equality), the asymptotic variance is simply the inverse of this matrix:

$$V^{IV} = \text{AVar} (\sqrt{N}(\hat{\beta}_{IV} - \beta)) = (E [\Pi' z_i' K' (K\Sigma(z_i)K')^{-1} Kz_i \Pi])^{-1} \quad (17)$$

Note that this estimator is completely unfeasible in practice because it requires knowledge of population parameters that are typically unobserved:

- **Unknown first stage:** It depends on the true structural parameter matrix Π linking z_i and x_i .
- **Unknown error covariance:** It requires exact knowledge of $\Omega(z_i) = K\Sigma(z_i)K'$. To construct $\Sigma(z_i)$, we need the true values of the AR(1) parameters ρ and σ^2 .

Efficiency Comparisons

We evaluate under which values of ρ the standard FD or WG 2SLS estimators match the optimal weighting matrix of $\hat{\beta}_{IV}$. Because efficiency in this class is invariant to the choice of the full-row rank transformation matrix, we will evaluate the optimal framework by setting $K = D$ (the first-difference operator) and observe how V^{IV} simplifies.

When is FD optimally efficient? If $\rho = 1$, the level error $v_{it} = v_{i,t-1} + \epsilon_{it}$ follows a random walk. Applying the first-difference matrix D yields $Dv_i = \Delta v_i = \epsilon_i$. Since ϵ_i is conditionally i.i.d., the conditional variance of the transformed errors is:

$$D\Sigma(z_i)D' = \sigma^2 I_{T-1} \quad (18)$$

Plugging $(D\Sigma(z_i)D')^{-1} = \frac{1}{\sigma^2} I_{T-1}$ into the formula for V^{IV} , the optimal weights matrix simplifies identically to the identity matrix, and the variance becomes:

$$V^{IV} = \left(E \left[\Pi' z_i' D' \left(\frac{1}{\sigma^2} I_{T-1} \right) D z_i \Pi \right] \right)^{-1} = \sigma^2 (E [\Pi' \Delta z_i' \Delta z_i \Pi])^{-1} \quad (19)$$

This perfectly matches the asymptotic variance of the standard FD 2SLS estimator, V^{FD} . The unfeasible estimator itself simplifies to:

$$\hat{\beta}_{IV} = \left(\sum_{i=1}^N \Pi' \Delta z_i' \Delta x_i \right)^{-1} \left(\sum_{i=1}^N \Pi' \Delta z_i' \Delta y_i \right) \quad (20)$$

Since the feasible 2SLS FD estimator replaces Π with a consistent first-stage estimate $\hat{\Pi}$, it is asymptotically equivalent. Thus, the variance terms perfectly align and $\hat{\beta}_{FD}$ is **efficient** when $\rho = 1$.

When is WG optimally efficient? If $\rho = 0$, the level errors $v_{it} = \epsilon_{it}$ are strictly classical (Assumption A2 from the lectures), meaning $\Sigma(z_i) = \sigma^2 I_T$. As shown in the course slides, first-differencing classical errors induces an MA(1)-type serial correlation:

$$D\Sigma(z_i)D' = \sigma^2 DD' \quad (21)$$

Plugging $(D\Sigma(z_i)D')^{-1} = \frac{1}{\sigma^2}(DD')^{-1}$ into the optimal IV formula, the estimator weights the matrices by $D'(DD')^{-1}D$. Using the key matrix identity connecting first-differences to the within-group (time-demeaning) operator Q :

$$D'(DD')^{-1}D = I_T - \frac{1}{T}\iota\iota' \equiv Q \quad (22)$$

Substituting this directly into the asymptotic variance formula V^{IV} , the matrix term collapses securely into Q :

$$V^{IV} = \left(E \left[\Pi' z_i' D' \left(\frac{1}{\sigma^2} (DD')^{-1} \right) D z_i \Pi \right] \right)^{-1} = \sigma^2 (E [\Pi' z_i' Q z_i \Pi])^{-1} = \sigma^2 (E [\Pi' \tilde{z}_i \tilde{z}_i' \Pi])^{-1} \quad (23)$$

This is exactly the asymptotic variance of the WG 2SLS estimator, V^{WG} . The unfeasible estimator similarly simplifies to projecting the within-transformed variables onto one another:

$$\hat{\beta}_{IV} = \left(\sum_{i=1}^N \Pi' z_i' Q x_i \right)^{-1} \left(\sum_{i=1}^N \Pi' z_i' Q y_i \right) = \left(\sum_{i=1}^N \Pi' \tilde{z}_i \tilde{x}_i \right)^{-1} \left(\sum_{i=1}^N \Pi' \tilde{z}_i \tilde{y}_i \right) \quad (24)$$

This matches the exact structure of $\hat{\beta}_{WG}$, which uses within-transformed instruments $\tilde{z}_i$. Thus, $\hat{\beta}_{WG}$ is **exactly equivalent to optimal GLS in first differences, and achieves maximum efficiency when $\rho = 0$** .

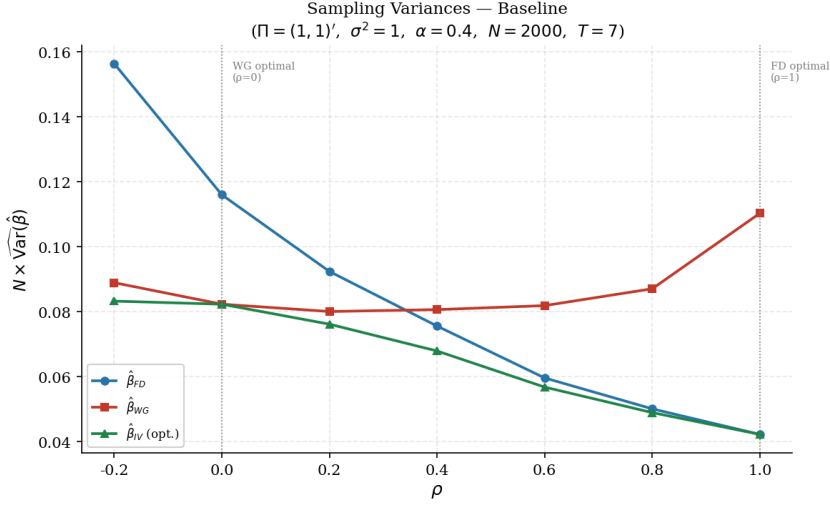
(c) Monte Carlo Simulation Results

Using the parameters specified ($N = 2000, T = 7, \beta = 1, r = 2, \Pi = (1, 1)', \sigma^2 = 1$, and $\alpha = 0.4$), we simulated the distributions of the First-Difference ($\hat{\beta}_{FD}$), Within-Groups ($\hat{\beta}_{WG}$), and the unfeasible optimal IV ($\hat{\beta}_{IV}$) estimators over $n_{MC} = 10,000$ iterations.

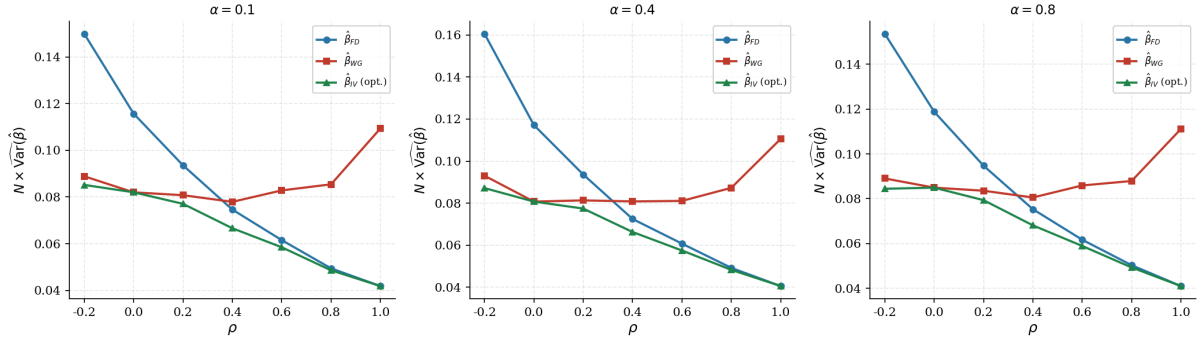
The baseline figure confirms the theoretical predictions established in part (b):

- **Optimal Lower Bound:** The unfeasible optimal IV estimator $\hat{\beta}_{IV}$ (green line) strictly dominates, providing the efficiency lower bound across all values of ρ .
- **WG Efficiency at $\rho = 0$:** The sampling variance of $\hat{\beta}_{WG}$ (red line) perfectly intersects with the optimal IV bound exactly at $\rho = 0$, confirming that Within-Groups is the optimally efficient transformation for i.i.d. classical errors.
- **FD Efficiency at $\rho = 1$:** The sampling variance of $\hat{\beta}_{FD}$ (blue line) decays monotonically as serial correlation increases, intersecting perfectly with the optimal IV bound at $\rho = 1$, confirming that First-Differences is optimally efficient for random-walk errors.

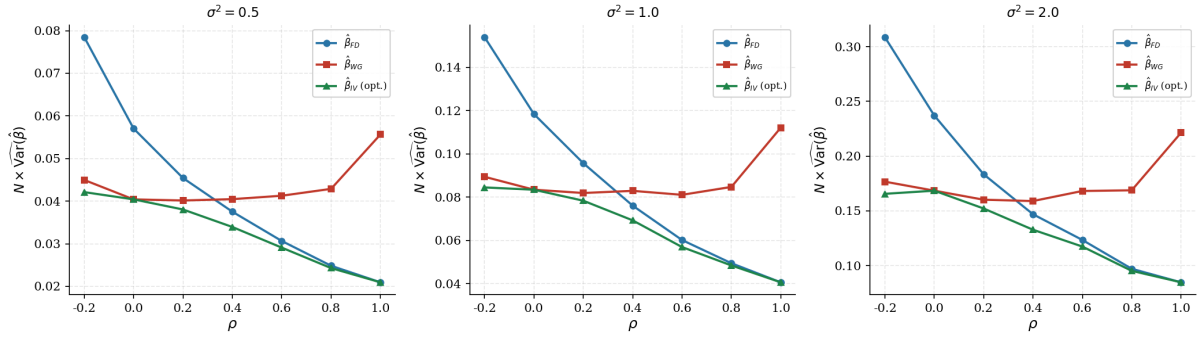
In the supplementary figures, we alter the three parameters, one at a time, over three different options. While altering these parameters scales the absolute magnitude of the variances up or down, the fundamental efficiency crossover points remain fixed. Regardless of α or σ^2 , WG remains uniquely optimal at $\rho = 0$ and FD remains uniquely optimal at $\rho = 1$.



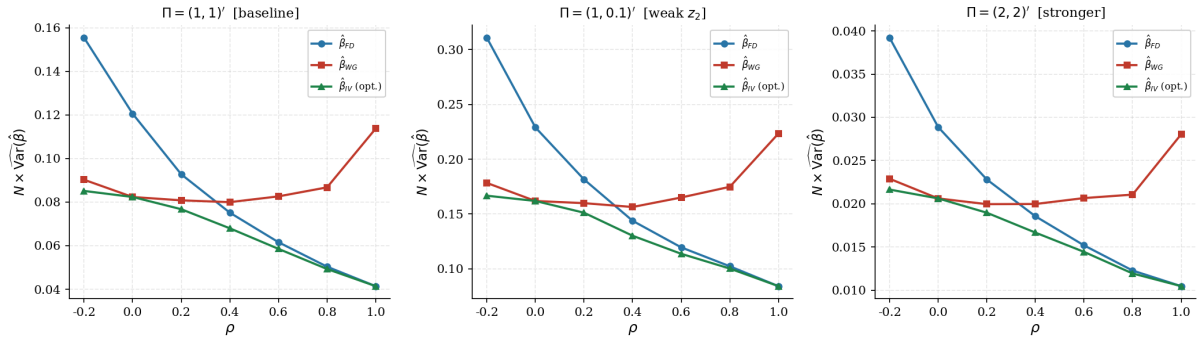
Effect of Endogeneity Strength α on Sampling Variances



Effect of Error Variance σ^2 on Sampling Variances



Effect of First-Stage Coefficient Π on Sampling Variances



(d) Main Lessons and Empirical Application

The main lesson from this exercise is that the relative efficiency of the First-Difference (FD) and Within-Group (WG) estimators is uniquely determined by the serial correlation structure of the idiosyncratic errors (v_{it}). While both estimators are consistent under the assumption of strict exogeneity, their finite-sample and asymptotic performance varies significantly depending on the persistence of the unobserved shocks, represented by the AR(1) parameter ρ :

- When $\rho = 0$ (classical i.i.d. errors), the **Within-Group estimator is optimally efficient**. This is because WG is mechanically equivalent to performing GLS on a differenced system, optimally correcting for the MA(1) serial correlation that differencing induces in i.i.d. errors.
- When $\rho = 1$ (errors follow a random walk), the **First-Difference estimator is optimally efficient**. In this case, differencing the random walk errors stabilizes them, producing conditionally spherical and i.i.d. transformed errors ($\Delta v_{it} = \epsilon_{it}$), making standard 2SLS in differences the optimal approach.

If the error process is somewhere between these two extremes ($0 < \rho < 1$), neither standard estimator is perfectly efficient, representing a trade-off. The unfeasible optimal IV (or a feasible GLS/GMM counterpart) provides the lowest possible variance.

In an applied empirical setting, an econometrician would choose the First-Difference estimator over the Within-Groups estimator if they have prior information, diagnostic evidence, or theoretical reasons suggesting that the idiosyncratic shocks are persistent. If the unobserved shocks v_{it} are suspected to follow a random walk or possess a unit root ($\rho \approx 1$), or the ρ is expected to be large, the FD transformation is superior. It stabilizes the variance and eliminates the permanent nature of the shocks, resulting in transformed errors that are nearly white noise, restoring efficiency.

So, it is always better to choose FD in the presence of *any* serial correlation? Note how in the previous charts the sampling variance of the Within-Groups estimator is relatively constant unless the process has a unit root. This means that, in some way, the “risk” of choosing a wrong assumption when choosing between both estimators is lower for WG compared to FD. If the econometrician chooses arbitrarily to estimate a FD because there is some serial correlation in the error term, there is a large risk of losing efficiency, specially if $\rho \leq 0.2$. WG is better (more efficient) when there is little to no serial correlation, and FD is better (more efficient) when there is a large serial correlation. When in doubt, it is always feasible to use a WG estimator and verify if the residuals are close to having a unit root. If that is the case, then a FD estimator is likely better.

Note that the sampling variance of the FD estimator is extremely high for low values of ρ and only becomes efficient as ρ approaches 1. Conversely, the WG estimator’s variance is relatively stable across a wide range of this parameter, only becoming highly inefficient at the extreme of a pure random walk.

This reveals that the risk of being wrong is asymmetric. Choosing FD when the true persistence is low (e.g., $\rho = 0.2$) results in a massive loss of efficiency compared to WG. In contrast, choosing WG when persistence is moderately high (e.g., $\rho = 0.8$) results in only a minor efficiency loss compared to FD. Therefore, the WG estimator can be seen as a more robust or “safer” default choice in the face of uncertainty.

A sound empirical strategy would be to use the WG estimator as a baseline and then perform a diagnostic test on its residuals. If these tests provide strong evidence that the errors contain a unit root, or any form of very strong serial correlation, then switching to the FD estimator is justified and likely to be more efficient.

In summary, the optimal choice depends on whether the original level errors behave more like “white noise” (favoring WG) or a “random walk” (favoring FD).

Exercise 2: Income and Consumption

(a) Deriving the Covariance Structure

To derive the expression for the unconditional covariance matrix $\Omega(\theta) = \text{Var}(s_i)$, we first need to explicitly characterize the processes for changes in income (Δy_{it}) and changes in consumption (Δc_{it}).

Model Transformation (First Differences)

We are given the levels equation for income:

$$y_{it} = \eta_{it} + \varepsilon_{it}$$

where $\eta_{it} = \eta_{i,t-1} + u_{it}$ and $\varepsilon_{it} = v_{it} + \psi_\varepsilon v_{i,t-1}$.

Taking the first difference of income yields:

$$\Delta y_{it} = (\eta_{it} - \eta_{i,t-1}) + (\varepsilon_{it} - \varepsilon_{i,t-1})$$

Substituting the assumed data-generating processes for η_{it} and ε_{it} :

$$\Delta y_{it} = u_{it} + (v_{it} + \psi_\varepsilon v_{i,t-1}) - (v_{i,t-1} + \psi_\varepsilon v_{i,t-2})$$

Grouping the terms by the timing of the shocks, we obtain an MA(2) representation for income growth:

$$\Delta y_{it} = u_{it} + v_{it} + (\psi_\varepsilon - 1)v_{i,t-1} - \psi_\varepsilon v_{i,t-2} \quad (25)$$

The equation for changes in consumption is already given directly in first differences as an MA(1)-type process:

$$\Delta c_{it} = \alpha u_{it} + \alpha \beta v_{it} + e_{it} + \psi_c e_{i,t-1} \quad (26)$$

Let $s_{it} = \begin{pmatrix} \Delta y_{it} \\ \Delta c_{it} \end{pmatrix}$ be the 2×1 vector of observables at time t . The full vector for individual i is $s_i = (s'_{i2}, s'_{i3}, \dots, s'_{iT})'$, which has dimension $2(T-1) \times 1$. Because the underlying innovations u_{it} , v_{it} , and e_{it} are mutually orthogonal and serially uncorrelated at all lags and leads, we can derive the covariance structure block by block using the Law of Iterated Expectations.

Contemporaneous Variance-Covariance Matrix (Lag 0)

Let $\Sigma_0 \equiv \text{Var}(s_{it}) = \text{E}[s_{it}s'_{it}]$. We evaluate the variances and the contemporaneous covariance:

- **Variance of Income Growth:**

$$\begin{aligned} \text{Var}(\Delta y_{it}) &= \text{E}[u_{it}^2] + \text{E}[v_{it}^2] + (\psi_\varepsilon - 1)^2 \text{E}[v_{i,t-1}^2] + (-\psi_\varepsilon)^2 \text{E}[v_{i,t-2}^2] \\ &= \sigma_\eta^2 + \sigma_\varepsilon^2 + (\psi_\varepsilon^2 - 2\psi_\varepsilon + 1)\sigma_\varepsilon^2 + \psi_\varepsilon^2 \sigma_\varepsilon^2 \\ &= \sigma_\eta^2 + 2(1 - \psi_\varepsilon + \psi_\varepsilon^2)\sigma_\varepsilon^2 \end{aligned}$$

- **Variance of Consumption Growth:**

$$\begin{aligned} \text{Var}(\Delta c_{it}) &= \alpha^2 \text{E}[u_{it}^2] + \alpha^2 \beta^2 \text{E}[v_{it}^2] + \text{E}[e_{it}^2] + \psi_c^2 \text{E}[e_{i,t-1}^2] \\ &= \alpha^2 \sigma_\eta^2 + \alpha^2 \beta^2 \sigma_\varepsilon^2 + (1 + \psi_c^2)\sigma_c^2 \end{aligned}$$

- **Contemporaneous Covariance:**

$$\text{Cov}(\Delta y_{it}, \Delta c_{it}) = \text{E} \left[\left(u_{it} + v_{it} + (\psi_\varepsilon - 1)v_{i,t-1} - \psi_\varepsilon v_{i,t-2} \right) \left(\alpha u_{it} + \alpha \beta v_{it} + e_{it} + \psi_c e_{i,t-1} \right) \right]$$

Due to orthogonality across different shocks and times, only the terms strictly corresponding to time t do not vanish:

$$\text{Cov}(\Delta y_{it}, \Delta c_{it}) = \alpha \text{E}[u_{it}^2] + \alpha \beta \text{E}[v_{it}^2] = \alpha \sigma_\eta^2 + \alpha \beta \sigma_\varepsilon^2$$

Thus, the 2×2 block matrix Σ_0 is:

$$\Sigma_0 = \begin{pmatrix} \sigma_\eta^2 + 2(1 - \psi_\varepsilon + \psi_\varepsilon^2)\sigma_\varepsilon^2 & \alpha \sigma_\eta^2 + \alpha \beta \sigma_\varepsilon^2 \\ \alpha \sigma_\eta^2 + \alpha \beta \sigma_\varepsilon^2 & \alpha^2 \sigma_\eta^2 + \alpha^2 \beta^2 \sigma_\varepsilon^2 + (1 + \psi_c^2)\sigma_c^2 \end{pmatrix} \quad (27)$$

First-Order Autocovariance Matrix (Lag 1)

Let $\Gamma_1 \equiv \text{Cov}(s_{it}, s_{i,t-1}) = \text{E}[s_{it}s'_{i,t-1}]$. We evaluate the cross-products between period t and $t - 1$:

- **Autocovariance of Income Growth:**

$$\begin{aligned} \text{Cov}(\Delta y_{it}, \Delta y_{i,t-1}) &= \text{E} \left[\left(\cdots + (\psi_\varepsilon - 1)v_{i,t-1} - \psi_\varepsilon v_{i,t-2} \right) \left(\cdots + v_{i,t-1} + (\psi_\varepsilon - 1)v_{i,t-2} - \cdots \right) \right] \\ &= (\psi_\varepsilon - 1)\sigma_\varepsilon^2 - \psi_\varepsilon(\psi_\varepsilon - 1)\sigma_\varepsilon^2 = -(1 - \psi_\varepsilon)^2\sigma_\varepsilon^2 \end{aligned}$$

- **Autocovariance of Consumption Growth:**

$$\text{Cov}(\Delta c_{it}, \Delta c_{i,t-1}) = \text{E} \left[\left(\cdots + \psi_c e_{i,t-1} \right) \left(\cdots + e_{i,t-1} + \cdots \right) \right] = \psi_c \sigma_c^2$$

- **Cross-Covariances (Income at t , Consumption at $t - 1$):**

$$\text{Cov}(\Delta y_{it}, \Delta c_{i,t-1}) = \text{E} \left[\left(\cdots + (\psi_\varepsilon - 1)v_{i,t-1} + \cdots \right) \left(\cdots + \alpha \beta v_{i,t-1} + \cdots \right) \right] = \alpha \beta (\psi_\varepsilon - 1)\sigma_\varepsilon^2$$

- **Cross-Covariances (Consumption at t , Income at $t - 1$):**

$$\text{Cov}(\Delta c_{it}, \Delta y_{i,t-1}) = 0$$

(Because Δc_{it} only contains elements driven by shocks at time t and measurement errors from $t - 1$, none of which overlap with the income shocks making up $\Delta y_{i,t-1}$)

Thus, the 2×2 lag-1 autocovariance matrix Γ_1 is:

$$\Gamma_1 = \begin{pmatrix} -(1 - \psi_\varepsilon)^2\sigma_\varepsilon^2 & \alpha \beta (\psi_\varepsilon - 1)\sigma_\varepsilon^2 \\ 0 & \psi_c \sigma_c^2 \end{pmatrix} \quad (28)$$

Second-Order Autocovariance Matrix (Lag 2)

Let $\Gamma_2 \equiv \text{Cov}(s_{it}, s_{i,t-2}) = \text{E}[s_{it}s'_{i,t-2}]$.

- **Autocovariance of Income Growth:**

$$\text{Cov}(\Delta y_{it}, \Delta y_{i,t-2}) = \text{E} \left[\left(\cdots - \psi_\varepsilon v_{i,t-2} \right) \left(\cdots + v_{i,t-2} + \cdots \right) \right] = -\psi_\varepsilon \sigma_\varepsilon^2$$

- **Autocovariance of Consumption Growth:**

$$\text{Cov}(\Delta c_{it}, \Delta c_{i,t-2}) = 0$$

- **Cross-Covariances:**

$$\text{Cov}(\Delta y_{it}, \Delta c_{i,t-2}) = \text{E} \left[\left(\cdots - \psi_\varepsilon v_{i,t-2} \right) \left(\cdots + \alpha \beta v_{i,t-2} + \cdots \right) \right] = -\alpha \beta \psi_\varepsilon \sigma_\varepsilon^2$$

$$\text{Cov}(\Delta c_{it}, \Delta y_{i,t-2}) = 0$$

Thus, the lag-2 autocovariance matrix Γ_2 is:

$$\Gamma_2 = \begin{pmatrix} -\psi_\varepsilon \sigma_\varepsilon^2 & -\alpha \beta \psi_\varepsilon \sigma_\varepsilon^2 \\ 0 & 0 \end{pmatrix} \quad (29)$$

Higher-Order Lags ($|t - s| \geq 3$)

Since income growth follows an MA(2) process and consumption growth follows an MA(1) process with independent measurement errors, all autocorrelations and cross-correlations vanish for lags strictly greater than 2. Thus, for any $k \geq 3$:

$$\Gamma_k = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad (30)$$

Full Covariance Matrix $\Omega(\theta)$

Collecting our results, the covariance matrix for the full sequence $s_i = (\Delta y_{i2}, \Delta c_{i2}, \dots, \Delta y_{iT}, \Delta c_{iT})'$ is a $2(T-1) \times 2(T-1)$ matrix.

Using the individual submatrices we evaluated above, we assemble the final structure of $\Omega(\theta)$ natively matching the auto-covariance profile:

$$\Omega(\theta) = \begin{pmatrix} \Sigma_0 & \Gamma'_1 & \Gamma'_2 & \mathbf{0} & \dots & \mathbf{0} \\ \Gamma_1 & \Sigma_0 & \Gamma'_1 & \Gamma'_2 & \dots & \mathbf{0} \\ \Gamma_2 & \Gamma_1 & \Sigma_0 & \Gamma'_1 & \dots & \mathbf{0} \\ \mathbf{0} & \Gamma_2 & \Gamma_1 & \Sigma_0 & \dots & \mathbf{0} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \dots & \Sigma_0 \end{pmatrix} \quad (31)$$

Where $\mathbf{0}$ defines a 2×2 zero matrix. Note that the main diagonal covers the covariance matrix between Δy_{it} and Δc_{it} , the first off-diagonal covers the first order autocovariance matrix between these two variables, and the second off-diagonal the second order autocovariance matrix. Every other component has a zero matrix on it.

(b) Estimation Strategy via GMM

To exploit the theoretical link derived in part (a), we can use a **Generalized Method of Moments (GMM)** approach. Because the structural parameters in $\theta = (\sigma_\eta^2, \psi_\varepsilon, \sigma_\varepsilon^2, \alpha, \beta, \psi_c, \sigma_c^2)$ are time-invariant, the theoretical covariance matrix $\Omega(\theta)$ implies that all the identifying information is contained entirely within the contemporaneous variance Σ_0 , the lag-1 autocovariance Γ_1 , and the lag-2 autocovariance Γ_2 .

Therefore, the problem is reduced to estimating these specific matrices from the data and finding the parameter vector θ that minimizes the distance between the theoretical moments and their sample counterparts.

Here is the step-by-step implementation strategy:

1. Construct the Sample Moments Assuming the data $s_{it} = (\Delta y_{it}, \Delta c_{it})'$ is mean-zero (or has been cross-sectionally demeaned to remove aggregate time fixed effects), we pool the observations over time to compute the sample variance and autocovariances. These form our empirical vector $\hat{m}$:

- **Sample Contemporaneous Variance ($\hat{\Sigma}_0$):**

$$\hat{\Sigma}_0 = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{T-1} \sum_{t=2}^T s_{it} s'_{it} \right)$$

- **Sample Lag-1 Autocovariance ($\hat{\Gamma}_1$):**

$$\hat{\Gamma}_1 = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{T-2} \sum_{t=3}^T s_{it} s'_{i,t-1} \right)$$

- **Sample Lag-2 Autocovariance ($\hat{\Gamma}_2$):**

$$\hat{\Gamma}_2 = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{T-3} \sum_{t=4}^T s_{it} s'_{i,t-2} \right)$$

2. Define the Theoretical Moment Conditions The theoretical moments $m(\theta)$, derived from the structure of Σ_0, Γ_1 , and Γ_2 , are given by the following 8 equations:

1. $\text{Var}(\Delta y_{it}) = \sigma_\eta^2 + 2(1 - \psi_\varepsilon + \psi_\varepsilon^2)\sigma_\varepsilon^2$
2. $\text{Var}(\Delta c_{it}) = \alpha^2 \sigma_\eta^2 + \alpha^2 \beta^2 \sigma_\varepsilon^2 + (1 + \psi_\varepsilon^2)\sigma_c^2$
3. $\text{Cov}(\Delta y_{it}, \Delta c_{it}) = \alpha \sigma_\eta^2 + \alpha \beta \sigma_\varepsilon^2$
4. $\text{Cov}(\Delta y_{it}, \Delta y_{i,t-1}) = -(1 - \psi_\varepsilon)^2 \sigma_\varepsilon^2$
5. $\text{Cov}(\Delta c_{it}, \Delta c_{i,t-1}) = \psi_c \sigma_c^2$
6. $\text{Cov}(\Delta y_{it}, \Delta c_{i,t-1}) = \alpha \beta (\psi_\varepsilon - 1) \sigma_\varepsilon^2$
7. $\text{Cov}(\Delta y_{it}, \Delta y_{i,t-2}) = -\psi_\varepsilon \sigma_\varepsilon^2$
8. $\text{Cov}(\Delta y_{it}, \Delta c_{i,t-2}) = -\alpha \beta \psi_\varepsilon \sigma_\varepsilon^2$

3. The Minimum Distance Objective Function We estimate θ by minimizing the quadratic distance between the empirical sample moments $\hat{m}$ and the theoretical moments $m(\theta)$, weighted by some positive-definite matrix W :

$$\hat{\theta} = \arg \min_{\theta} (\hat{m} - m(\theta))' W (\hat{m} - m(\theta))$$

For the weighting matrix W , we have two standard choices:

- **Equally Weighted Minimum Distance:** We can simply set $W = I$ (the identity matrix). This gives equal weight to all moments and is often preferred in panel data covariance modeling to avoid the small-sample bias introduced by estimating an optimal weight matrix.
- **Two-Step Optimal GMM:** We can obtain an initial estimate using $W = I$, use it to estimate the variance-covariance matrix of the sample moments $\hat{V}$, and then re-run the minimization using the optimal weight matrix $W = \hat{V}^{-1}$ to achieve asymptotic efficiency.

4. Parameter Identification By setting up the problem this way, the optimization routine naturally exploits the structure of the submatrices to uniquely identify the parameters:

- **From Γ_1 and Γ_2 :** The ratio of $\text{Cov}(\Delta y_{it}, \Delta y_{i,t-2})$ to $\text{Cov}(\Delta y_{it}, \Delta y_{i,t-1})$ identifies ψ_ε and σ_ε^2 .
- **From Σ_0 :** Knowing ψ_ε and σ_ε^2 , the variance $\text{Var}(\Delta y_{it})$ directly identifies σ_η^2 .
- **From the cross-covariances:** $\text{Cov}(\Delta y_{it}, \Delta c_{i,t-2})$ identifies the compound parameter $\alpha\beta$. Plugging this into $\text{Cov}(\Delta y_{it}, \Delta c_{it})$ identifies α , which subsequently separates β .
- **From the consumption block:** Knowing the income and response parameters leaves two equations ($\text{Var}(\Delta c_{it})$ and $\text{Cov}(\Delta c_{it}, \Delta c_{i,t-1})$) to solve for the two remaining measurement error parameters ψ_c and σ_c^2 .

(c) Income Process Estimates and Covariance Fit

Estimates and Standard Errors

Using a Minimum Distance (GMM) estimator on the 12 income-only moments (5 variances, 4 lag-1 covariances, and 3 lag-2 covariances), we obtain the following parameter estimates for the income process:

Parameter	Coefficient	Std. Err.	z	P-Value	[95% Conf. Interval]	
σ_η^2	0.0372688	0.0060073	6.20	0.000	0.0254947	0.0490428
ψ_ε	0.0059951	0.0775638	0.08	0.938	-0.1460271	0.1580174
σ_ε^2	0.0497815	0.0067707	7.35	0.000	0.0365112	0.0630518

Both the variance of the permanent shock (σ_η^2) and the variance of the transitory shock (σ_ε^2) are highly statistically significant ($p < 0.001$). However, the MA(1) parameter for the transitory component (ψ_ε) is economically very small and statistically indistinguishable from zero by a large margin ($p = 0.938$).

What do we learn about $\sigma_\eta^2/\sigma_\varepsilon^2$?

Using the point estimates, the relative proportion of permanent to transitory shocks is:

$$\frac{\hat{\sigma}_\eta^2}{\hat{\sigma}_\varepsilon^2} = \frac{0.0372688}{0.0497815} \approx 0.7486$$

This ratio is approximately 0.75, meaning that the variance of the permanent shock is roughly three-quarters the size of the variance of the transitory shock. From this, we learn that while both components are crucial drivers of income volatility, transitory shocks dominate and generate slightly more volatility in the data than permanent shocks. Furthermore, because $\hat{\psi}_\varepsilon \approx 0$, the transitory component is essentially an i.i.d. process rather than a persistent moving average.

Heuristic Comparison of Covariance Matrices

To heuristically evaluate the model's fit, we compare the theoretical predictions of the MA(2) model to the empirical covariance matrix of income growth from the data:

Alignments with the theoretical model:

- **Lag 1 Covariance (First off-diagonal):** The model predicts a distinctly negative lag-1 covariance: $-(1 - \hat{\psi}_\varepsilon)^2 \hat{\sigma}_\varepsilon^2 \approx -0.050$. The empirical matrix shows consistently negative lag-1 values ($-0.040, -0.099, -0.058, -0.027$), which strongly supports the model's structure.

	dy2	dy3	dy4	dy5	dy6
dy2	0.126538				
dy3	-0.040414	0.181423			
dy4	-0.001069	-0.099277	0.189446		
dy5	-0.003720	0.003674	-0.058475	0.152093	
dy6	0.003677	0.002795	-0.002035	-0.026906	0.165880

- **Lag 2 Covariance (Second off-diagonal):** The model predicts $-\hat{\psi}_\varepsilon \hat{\sigma}_\varepsilon^2$. Since $\hat{\psi}_\varepsilon \approx 0$, this should be near zero. The empirical matrix shows exactly this: $-0.001, 0.004, -0.002$.

Observed Deviations and Suggested Improvements:

- **Non-stationarity of Variances:** The theoretical model imposes strict stationarity, meaning the variance (the main diagonal) should be perfectly constant across all years. The empirical matrix shows variances fluctuating from 0.126 up to 0.189. *Improvement:* To improve fit, the model specification could be relaxed to allow for time-varying variances of the shocks ($\sigma_{\eta,t}^2$ and $\sigma_{\varepsilon,t}^2$), reflecting aggregate macroeconomic conditions.
- **Higher-Order Lags (Lag 3+):** The strict MA(2) structure forces covariances at lags 3 and 4 to be exactly zero. While the empirical matrix shows very small numbers for these lags (e.g., $-0.0037, 0.0036$), they do not perfectly vanish. *Improvement:* Modeling the persistent shock η_{it} as an AR(1) process with a coefficient close to, but slightly less than 1 (e.g., $\rho \approx 0.95$), or modeling the transitory shock ε_{it} as an ARMA process, would allow these correlations to decay smoothly over time rather than cutting off abruptly.

(d) GMM Estimation and the Permanent Income Hypothesis

Estimates of the Full Parameter Vector (θ)

Using the full joint covariance structure of income and consumption growth (incorporating 33 moments total: income, consumption, and their cross-covariances), we estimate the full parameter vector θ . The GMM estimation yields the following results:

Parameter	Coefficient	Std. Err.	z	P-Value	[95% Conf. Interval]	
σ_η^2	0.0369	0.0057	6.40	0.000	0.0256	0.0482
ψ_ε	0.0093	0.0744	0.13	0.900	-0.1365	0.1551
σ_ε^2	0.0484	0.0065	7.34	0.000	0.0354	0.0613
α	0.2392	0.0587	4.07	0.000	0.1241	0.3543
β	0.3128	0.1834	1.70	0.088	-0.0467	0.6725
ψ_c	-0.4472	0.0299	-14.92	0.000	-0.5060	-0.3885
σ_c^2	0.0630	0.0022	28.40	0.000	0.0587	0.0674

Note that the estimates for the income process ($\sigma_\eta^2, \psi_\varepsilon, \sigma_\varepsilon^2$) remain highly consistent with the income-only estimates derived in Part (c), indicating that the system is stable.

Permanent Income Hypothesis

The strict baseline Permanent Income Hypothesis (PIH)—assuming quadratic utility, rational expectations, and perfect credit markets—generates two stark theoretical predictions regarding how consumption should respond to income shocks:

1. Consumers should fully adjust their consumption one-for-one in response to permanent shocks. Therefore, the theory predicts $\alpha = 1$.

2. Consumers should use borrowing and saving to perfectly smooth over temporary income fluctuations. Therefore, consumption should not respond to transitory shocks, predicting $\alpha\beta = 0$ (and since $\alpha \neq 0$, this requires $\beta = 0$).

Based on the GMM estimates of the Hall-Mishkin framework, we can draw the following conclusions:

Rejection of $\alpha = 1$

The estimated coefficient governing the response to permanent shocks is $\hat{\alpha} = 0.239$. This estimate is highly statistically significant ($p < 0.001$), but it is far below the theoretical prediction of 1. Households in this sample only translate roughly 24% of a permanent income shock into consumption. It implies that the PIH fails to capture reality, likely because households possess partial insurance mechanisms (e.g., family networks, government transfers, progressive taxation) or hold precautionary savings that blunt the impact of permanent shocks.

Rejection of $\beta = 0$

The parameter β scales the response of consumption to transitory shocks. Our estimate is $\hat{\beta} = 0.313$. While less precisely estimated, it is marginally significant ($p = 0.088$, significant at the 10% level) and economically meaningful. Because the total response to a transitory shock is modeled as $\alpha\beta$, households are adjusting their consumption by $\approx 0.239 \times 0.313 \approx 0.075$ (or roughly 7.5%) of the transitory shock.

This constitutes a rejection of the PIH prediction that transitory shocks should have no effect on consumption. This can be interpreted better in the presence of liquidity constraints; a subset of consumers cannot borrow to smooth consumption during temporary negative income shocks, forcing them to adjust consumption downward.

Exercise 3: Correlated Random Coefficients

(a) FD Estimator under Heterogeneous Slopes

Consider the random coefficients model with strictly exogenous regressors:

$$y_{it} = \alpha_i + \beta_i x_{it} + \gamma' z_{it} + v_{it} \quad (32)$$

where $w_{it} = (x_{it}, z'_{it})'$ is a $k_w \times 1$ vector of covariates, and we are given the assumption of **strict exogeneity**:

$$E[v_{it} \mid \beta_i, w_i] = E[v_{it} \mid \beta_i, x_{i1}, z_{i1}, \dots, x_{iT}, z_{iT}] = 0 \quad (33)$$

To eliminate the time-invariant unobserved heterogeneity α_i , we apply the first-difference operator. The transformed equation becomes:

$$\Delta y_{it} = \beta_i \Delta x_{it} + \gamma' \Delta z_{it} + \Delta v_{it} \quad (34)$$

Estimator and Asymptotic Limits

The researcher ignores the heterogeneity in the slope β_i and estimates a standard pooled OLS regression in first differences to obtain $(\hat{\beta}, \hat{\gamma})'$. Noting that first differences are only defined for $t = 2, \dots, T$, we can write the estimator as:

$$\begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \left(\sum_{i=1}^N \sum_{t=2}^T \Delta w_{it} \Delta w'_{it} \right)^{-1} \sum_{i=1}^N \sum_{t=2}^T \Delta w_{it} \Delta y_{it} \quad (35)$$

To find the asymptotic distribution as $N \rightarrow \infty$ with T fixed, we multiply by $1/N$ to obtain sample averages and substitute the true differenced model (34) into the estimator:

$$\begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \left(\frac{1}{N} \sum_{i=1}^N \sum_{t=2}^T \Delta w_{it} \Delta w'_{it} \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \sum_{t=2}^T \Delta w_{it} (\beta_i \Delta x_{it} + \gamma' \Delta z_{it} + \Delta v_{it}) \right)$$

By the LLN, as $N \rightarrow \infty$, the sample moments converge to their population counterparts. Additionally, by the strict exogeneity assumption (33), the error term is orthogonal to the regressors: $E[\Delta w_{it} \Delta v_{it}] = 0$. Thus, the probability limit is:

$$\text{plim}_{N \rightarrow \infty} \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \left(E \left[\sum_{t=2}^T \Delta w_{it} \Delta w'_{it} \right] \right)^{-1} E \left[\sum_{t=2}^T \Delta w_{it} (\beta_i \Delta x_{it} + \gamma' \Delta z_{it}) \right] \quad (36)$$

To compare this limit to the true mean parameter $\bar{\beta} = E[\beta_i]$ and the common parameter γ , we decompose the random slope into its population mean and an individual-specific deviation: $\beta_i = \bar{\beta} + (\beta_i - \bar{\beta})$. Substituting this into the expectation term yields:

$$\begin{aligned} & E \left[\sum_{t=2}^T \Delta w_{it} ((\bar{\beta} + (\beta_i - \bar{\beta})) \Delta x_{it} + \gamma' \Delta z_{it}) \right] \\ &= E \left[\sum_{t=2}^T \Delta w_{it} (\bar{\beta} \Delta x_{it} + \gamma' \Delta z_{it}) \right] + E \left[\sum_{t=2}^T \Delta w_{it} \Delta x_{it} (\beta_i - \bar{\beta}) \right] \\ &= E \left[\sum_{t=2}^T \Delta w_{it} \Delta w'_{it} \right] \begin{pmatrix} \bar{\beta} \\ \gamma \end{pmatrix} + E \left[(\beta_i - \bar{\beta}) \sum_{t=2}^T \Delta w_{it} \Delta x_{it} \right] \end{aligned}$$

Pre-multiplying this result by the inverse of the expected Hessian $\left(E \left[\sum_{t=2}^T \Delta w_{it} \Delta w'_{it} \right]\right)^{-1}$, we arrive at the final probability limit:

$$\text{plim}_{N \rightarrow \infty} \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \begin{pmatrix} \bar{\beta} \\ \gamma \end{pmatrix} + \left(E \left[\sum_{t=2}^T \Delta w_{it} \Delta w'_{it} \right] \right)^{-1} E \left[(\beta_i - \bar{\beta}) \sum_{t=2}^T \Delta w_{it} \Delta x_{it} \right] \quad (37)$$

Consistency with Heterogeneous Slopes

Equation (37) characterizes the asymptotic bias of the standard FD estimator in the presence of heterogeneous slopes. We evaluate the consistency under two distinct assumptions:

1. General Case (Fixed-Effects Perspective):

If $E[\beta_i | w_i]$ is left unrestricted, the deviation of the slope from its mean $(\beta_i - \bar{\beta})$ can be arbitrarily correlated with the differenced regressors $\Delta w_{it} \Delta x_{it}$. Consequently, the bias term $E \left[(\beta_i - \bar{\beta}) \sum_{t=2}^T \Delta w_{it} \Delta x_{it} \right] \neq 0$. The standard FD estimator is **inconsistent** for both the mean slope $\bar{\beta}$ and the common parameters γ , as the heterogeneity in β_i bleeds into the estimation of γ through the inverse matrix.

2. Random-Effects Perspective ($E[\beta_i | w_i] = \bar{\beta}$):

If we impose a “random effects” style assumption where the random coefficients are conditionally independent of the regressors, the conditional expectation of the deviation is zero: $E[\beta_i - \bar{\beta} | w_i] = 0$. By the LIE, we can unpack the unconditional expectation of the bias term:

$$E \left[(\beta_i - \bar{\beta}) \sum_{t=2}^T \Delta w_{it} \Delta x_{it} \right] = E \left[E[\beta_i - \bar{\beta} | w_i] \sum_{t=2}^T \Delta w_{it} \Delta x_{it} \right] = E \left[0 \cdot \sum_{t=2}^T \Delta w_{it} \Delta x_{it} \right] = 0 \quad (38)$$

Substituting this back into Equation (37) eliminates the second term entirely. Therefore, under the assumption that $E[\beta_i | w_i] = \bar{\beta}$, the standard first-difference estimator is **consistent**:

$$\hat{\beta} \xrightarrow{p} \bar{\beta} \quad \text{and} \quad \hat{\gamma} \xrightarrow{p} \gamma \quad (39)$$

(b) Test for Correlated Random Slopes

Determining $g(w_i)$

From our result in part (a), the asymptotic probability limit of the FD estimator is:

$$\text{plim}_{N \rightarrow \infty} \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \begin{pmatrix} \bar{\beta} \\ \gamma \end{pmatrix} + \left(E \left[\sum_{t=2}^T \Delta w_{it} \Delta w'_{it} \right] \right)^{-1} E \left[(\beta_i - \bar{\beta}) \sum_{t=2}^T \Delta w_{it} \Delta x_{it} \right] \quad (40)$$

We are looking for a $(k+1) \times 1$ function $g(w_i)$ such that $\text{Cov}(\beta_i, g(w_i)) = 0$ implies consistency of the estimator. Note that the covariance between the random slope β_i and any function of the data $g(w_i)$ can be written as:

$$\begin{aligned} \text{Cov}(\beta_i, g(w_i)) &= E[(\beta_i - E[\beta_i])(g(w_i) - E[g(w_i)])] \\ &= E[(\beta_i - E[\beta_i])g(w_i)] - (E[\beta_i] - E[\beta_i])E[g(w_i)] \\ &= E[(\beta_i - \bar{\beta})g(w_i)] \end{aligned}$$

By directly comparing this covariance expression to the bias term in Equation (40), we can easily identify $g(w_i)$. If we define:

$$g(w_i) = \sum_{t=2}^T \Delta w_{it} \Delta x_{it} \quad (41)$$

then the condition $\text{Cov}(\beta_i, g(w_i)) = 0$ exactly means $E[(\beta_i - \bar{\beta}) \sum_{t=2}^T \Delta w_{it} \Delta x_{it}] = 0$. This nullifies the bias term in (40), implying $\hat{\beta} \xrightarrow{p} \bar{\beta}$ and $\hat{\gamma} \xrightarrow{p} \gamma$. Thus, $g(w_i) = \sum_{t=2}^T \Delta w_{it} \Delta x_{it}$ is the sufficient summary statistic of the covariates that drives the inconsistency.

Deriving the Test

We now consider a linear projection model for the conditional expectation of the random slope:

$$E[\beta_i | w_i] = \lambda_0 + \lambda'_1 g(w_i) \quad (42)$$

Under the null hypothesis $H_0 : \lambda_1 = 0_{(k+1) \times 1}$, the conditional mean of the slope does not depend on $g(w_i)$, which implies $\text{Cov}(\beta_i, g(w_i)) = 0$, and thus the standard FD estimator is consistent for the mean slope.

To derive a test for H_0 , we decompose the random slope into its conditional expectation and a residual unobserved heterogeneity term η_i :

$$\beta_i = \lambda_0 + \lambda'_1 g(w_i) + \eta_i$$

where $E[\eta_i | w_i] = 0$ by construction.

We substitute this expression for β_i back into the original first-differenced equation:

$$\begin{aligned} \Delta y_{it} &= \beta_i \Delta x_{it} + \gamma' \Delta z_{it} + \Delta v_{it} \\ &= (\lambda_0 + \lambda'_1 g(w_i) + \eta_i) \Delta x_{it} + \gamma' \Delta z_{it} + \Delta v_{it} \\ &= \lambda_0 \Delta x_{it} + \gamma' \Delta z_{it} + \lambda'_1 (g(w_i) \Delta x_{it}) + (\eta_i \Delta x_{it} + \Delta v_{it}) \end{aligned} \quad (43)$$

Let $u_{it} = \eta_i \Delta x_{it} + \Delta v_{it}$ be the composite error term of this augmented regression. We can verify that this new error term is conditionally mean-zero given the covariates:

$$E[u_{it} | w_i] = E[\eta_i \Delta x_{it} + \Delta v_{it} | w_i] = \Delta x_{it} E[\eta_i | w_i] + E[\Delta v_{it} | w_i] = 0 + 0 = 0$$

Because $E[u_{it} | w_i] = 0$, the composite error is strictly exogenous with respect to the regressors. Therefore, estimating Equation (43) by pooled OLS will yield consistent estimates of λ_0, γ , and λ_1 .

Detailed Implementation

To implement this test in practice as a Variable Addition Test, the econometrician should follow these steps:

1. **Construct $g(w_i)$:** For each individual i , compute the $(k+1) \times 1$ vector $g(w_i) = \sum_{s=2}^T \Delta w_{is} \Delta x_{is}$. Note that this value aggregates over all time periods and is time-invariant for a given individual i .
2. **Generate the Auxiliary Regressors:** For each individual i and time period $t \geq 2$, generate the interaction term $r_{it} = g(w_i) \Delta x_{it}$. This r_{it} is a $(k+1) \times 1$ vector of observable augmented regressors.
3. **Run the Augmented Regression:** Estimate the following equation using pooled OLS over all N units and $T - 1$ periods:

$$\Delta y_{it} = \lambda_0 \Delta x_{it} + \gamma' \Delta z_{it} + \lambda'_1 r_{it} + u_{it}$$

which is mechanically equivalent to regressing Δy_{it} on Δw_{it} and r_{it} .

4. **Compute Cluster-Robust Standard Errors:** The composite error $u_{it} = \eta_i \Delta x_{it} + \Delta v_{it}$ is heteroskedastic and serially correlated within each individual i (due to both Δv_{it} being differenced and the unobserved heterogeneity η_i acting as an interactive random effect). Standard errors *must* be clustered at the individual i level to ensure valid inference.
5. **Perform the Wald Test:** Extract the estimated vector $\hat{\lambda}_1$ and its corresponding block from the cluster-robust variance-covariance matrix, $\widehat{\text{Var}}(\hat{\lambda}_1)$. Compute the Wald statistic:

$$W = \hat{\lambda}_1' \left(\widehat{\text{Var}}(\hat{\lambda}_1) \right)^{-1} \hat{\lambda}_1$$

Under the null hypothesis $H_0 : \lambda_1 = 0_{(k+1) \times 1}$, the statistic W follows an asymptotic χ^2 distribution with $k + 1$ degrees of freedom. Reject H_0 if W exceeds the critical value. If H_0 is rejected, there is statistical evidence of correlated random slopes, confirming the standard FD estimator is inconsistent.

(c) Test for Correlated Random Slopes

Implementation of the Test and Evidence of Correlation

To test for correlated random slopes, we implement the Test derived in part (b). The null hypothesis is $H_0 : \lambda_{1z} = 0$, meaning the conditional mean of the heterogeneous slope β_i does not depend on age squared, and therefore the standard FD estimator is consistent.

Based on the empirical output for the augmented regression, the coefficient on r_{zit} is 0.0001602. The Wald test for this coefficient yields:

- $F(1, 791) = 2.73$ ¹
- $p\text{-value} = 0.0990$

We reject the null hypothesis at the 10% significance level. There is marginal, but statistically meaningful evidence that the heterogeneous income trend (β_i) is correlated with age squared. Because the random slope is correlated with the regressors, the standard pooled First-Difference estimator is inconsistent, and the heterogeneity “bleeds” into the estimation of the common parameters.

Estimates of the Mean Slope (Test-Supported Approach)

Because we rejected H_0 , we must estimate the unconditional mean slope $\mathbb{E}[\beta_i]$ using the linear projection model supported by the test:

$$\mathbb{E}[\beta_i] = \lambda_0 + \lambda_{1x} \bar{g}_x + \lambda_{1z} \bar{g}_z$$

Because $\Delta x_{it} = 1$ (time changes by 1 each period), $g_x(w_i) = 5$ for all individuals. In our specification, the auxiliary variable r_{xit} perfectly absorbs the $\lambda_0 + \lambda_{1x} \bar{g}_x$ components. Therefore, the base intercept is $5 \times \hat{\lambda}_{1x} = 5 \times 0.041588 = 0.20794$.

We calculate the mean slope by adding the effect of age squared, evaluated at the sample mean of g_z ($\bar{g}_z = 894.14$):

$$\hat{\mathbb{E}}[\beta_i]_{\text{Test}} = 0.20794 + (0.0001602 \times 894.14) \approx \mathbf{0.3512}$$

¹Note that the degrees of freedom in this case is only 1, because $r_{xit} = 5$ for every agent and time period, so it's perfectly collinear to the intercept of the model.

Comparison to a Non Parametric Approach

The procedure discussed in class for estimating the mean of heterogeneous coefficients is the Generalized Within (Individual Averaging) approach. This method estimates the slope for each individual separately and averages them: $\hat{\beta} = \frac{1}{N} \sum \hat{\beta}_i$.

Because $\Delta x_{it} = 1$, estimating a heterogeneous slope for every individual in the differenced equation is mechanically identical to estimating an individual fixed effect on the differenced data. The intercept of this fixed-effects regression provides the exact sample average of these individual slopes. However, the class procedure is, in general terms, a superior method for estimation, as it requires a less rigid set of assumptions. The approach in this exercise forces a parametric assumption—that the slope depends *strictly linearly* on $g(w_i)$. The class procedure places zero restrictions on the distribution of β_i given the regressors and allows for full flexibility. Therefore, while the augmented regression is an excellent diagnostic test to detect the presence of correlation, the Generalized Within approach is the most robust way to actually estimate the distribution of the individual effects.

The fundamental difference in how each method models heterogeneity is illustrated in Figure 1. The parametric model forces all heterogeneity into a rigid, artificially narrow distribution, whereas the non-parametric approach reveals a more natural and dispersed pattern.

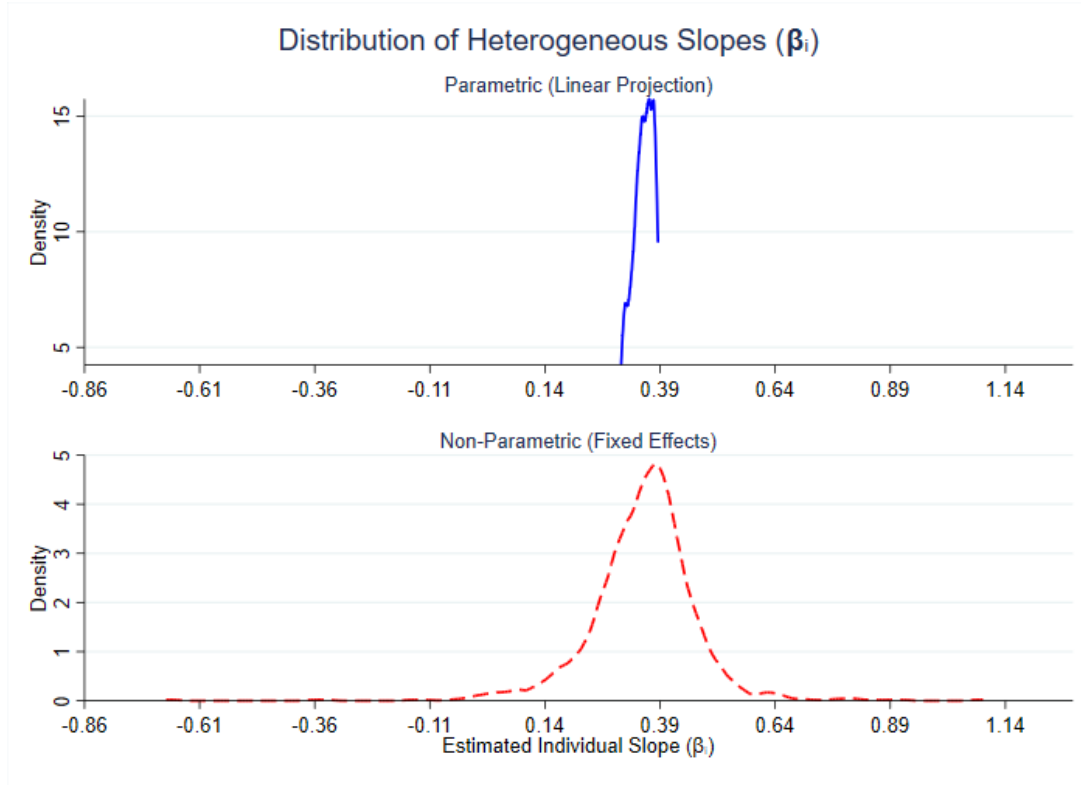


Figure 1: Distribution of Estimated Individual Slopes (β_i). The top panel displays the distribution from the parametric linear projection model, which forces all heterogeneity to be a rigid function of covariates. The bottom panel shows the fully non-parametric distribution from the Fixed Effects approach, revealing a more natural and dispersed pattern of heterogeneity. Both distributions are centered on the same mean (≈ 0.35).